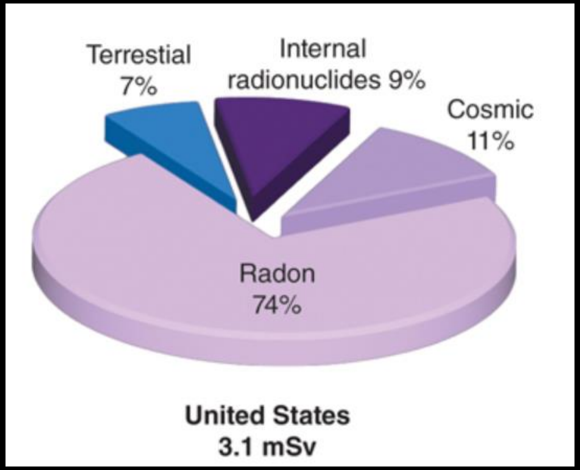
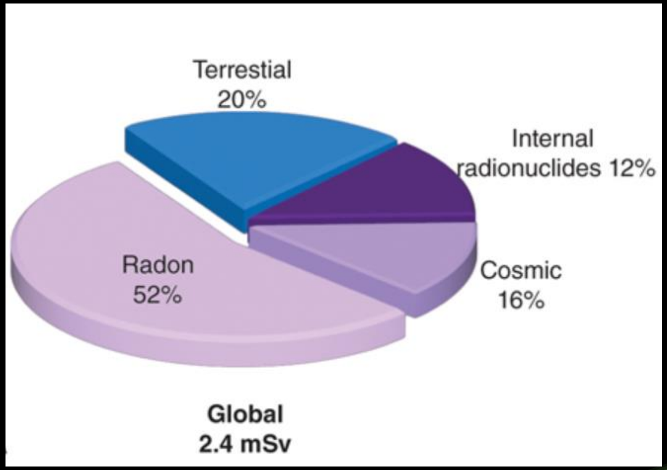
Radiation Safety and Protection

Source of radiation exposure

- Natural Background
  - Radon and its progeny
  - Space radiation
  - Internal Radionuclides
  - Terrestrial Radiation





Radon and its progeny

- Radon-222 is a radioactive element produced as an intermediate step in the decay of uranium-238.
- Its a gas, is released from the ground and enters homes and buildings.
- It is principal contributor to background radiation exposure and accounts for approximately 52%–74% of background radiation exposure.
- Radon and its decay products may become attached to dust particles that can be inhaled and deposited on the bronchial epithelium in the respiratory tract.
- It is the second most frequent cause of lung cancer and is estimated to cause 3%–14% of all lung cancers worldwide.

Cosmic radiation

- Cosmic rays, or space radiation, comprise protons, helium nuclei, and nuclei of heavier elements, as well as other particles generated by the interactions of primary space radiation with the Earth’s atmosphere.
- Exposure from cosmic radiation is primarily a function of altitude, almost doubling with each 2000-m increase in elevation because less atmosphere is present to attenuate the radiation.
- At sea level, the exposure from cosmic radiation is approximately 0.33 mSv/year; at an elevation of 1600 m (approximately 1 mile, the elevation of Denver, Colorado), it is approximately 0.50 mSv/year.
- Air travel exposes passengers and crew to cosmic radiation. The magnitude of exposure varies with flight altitude and duration. Cosmic radiation contributes approximately 11%–16% of background exposure.

Internal radionuclides

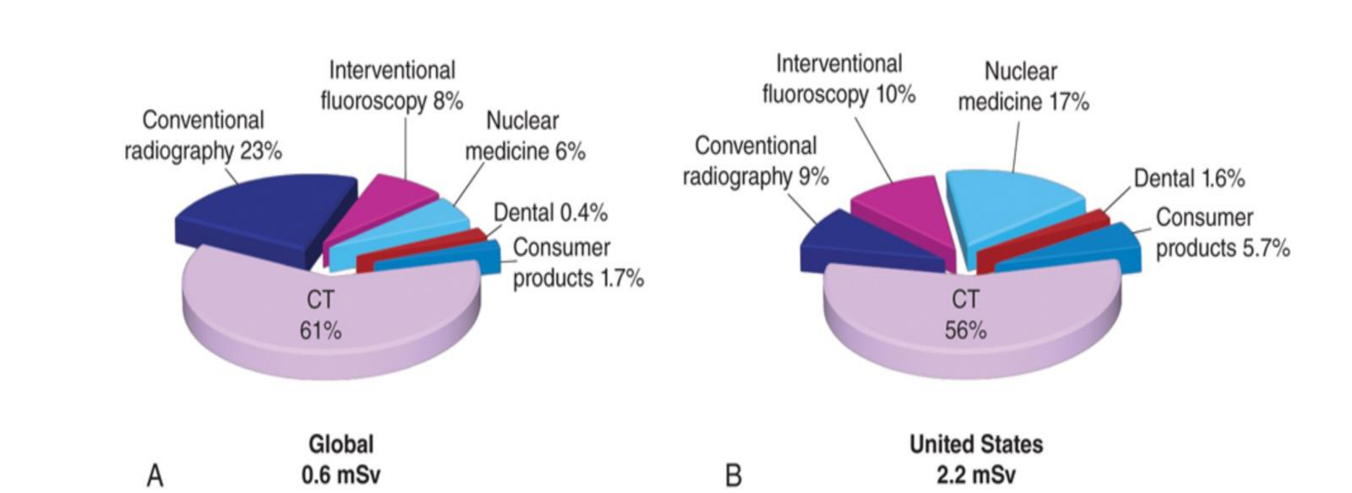
- Another source of background radiation is radionuclides that are ingested.
- The greatest internal exposure comes from foods containing uranium and thorium and their decay products, primarily potassium-40 but also rubidium-87, carbon-14, tritium, and others.
- The total exposure from ingestion contributes approximately 9%–12% of background exposure.

Terrestrial radiation

- Terrestrial radiation exposure is derived from radioactive nuclides in the soil, primarily potassium-40 and the radioactive decay products of uranium-238 and thorium-232.
- Most of the γ-radiation from these sources comes from the top 20 cm of soil.
- Indoor exposure from radionuclides is close to the exposure occurring outdoors because the shielding provided by structural materials balances the exposure from radioactive nuclides contained within these shielding materials.
- Terrestrial exposure contributes approximately 7%–20% of background exposure.

Source of radiation exposure

- Medical Sources
  - MDCT
  - Nuclear medicine
  - Interventional radiotherapy
  - Conventional radiography
  - Consumer products
  - Dental

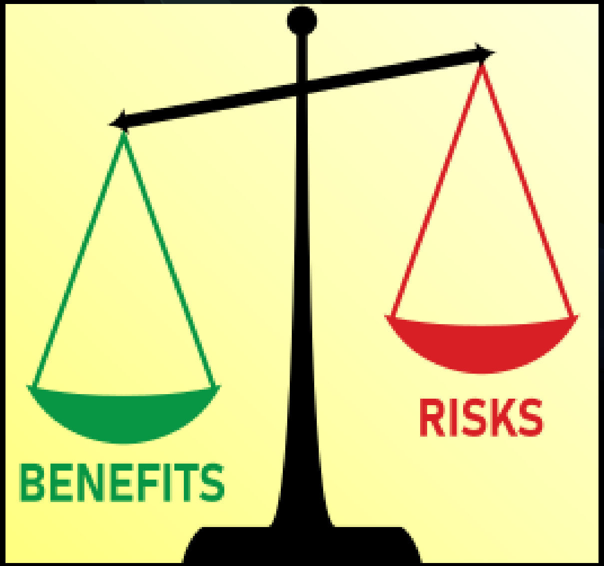


Dentomaxillofacial Radiology: Risks and Doses

‘Benefit to the patient far outweighs the radiation associated risks’

Doses from DMFR examinations must be:

- Optimized to produce a diagnostically acceptable image
- Less than the threshold needed to cause any deterministic effects
- Minimized to keep the risk of stochastic effects within an acceptable range

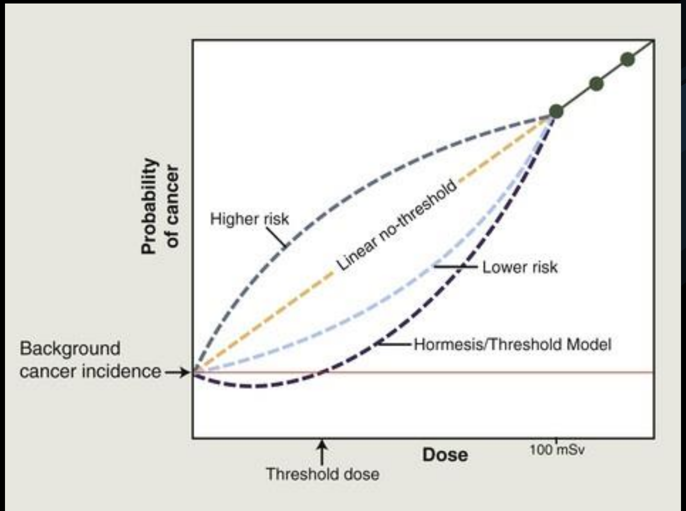


Estimating Cancer Risk From Diagnostic DMFR

The current paradigm of radiation protection is based on the linear no- threshold (**LNT**) hypothesis

- Predicts that there is a linear relationship between dose and the risk of inducing a new cancer, even at very low doses
- There is no threshold or “safe dose” below which there is no added cancer risk
- Complex damage to DNA, the basis of cancer formation, may occur with even one x-ray photon

Low-Dose Cancer Risk



- Doses of radiation higher than approximately 100 mSv result in a dose-dependent increase in the cancer rate.
- LNT posits that at doses less than 100 mSv, there is a linear relationship between dose and risk and that there is no threshold dose below which there is no additional risk.
- Alternate models propose that risk may be higher or lower than those predicted by the LNT model and that low doses may have a protective effect with a threshold dose. The LNT model is currently accepted as the approach to develop radiation protection guidelines.

Patient doses from Diagnostic DMFR

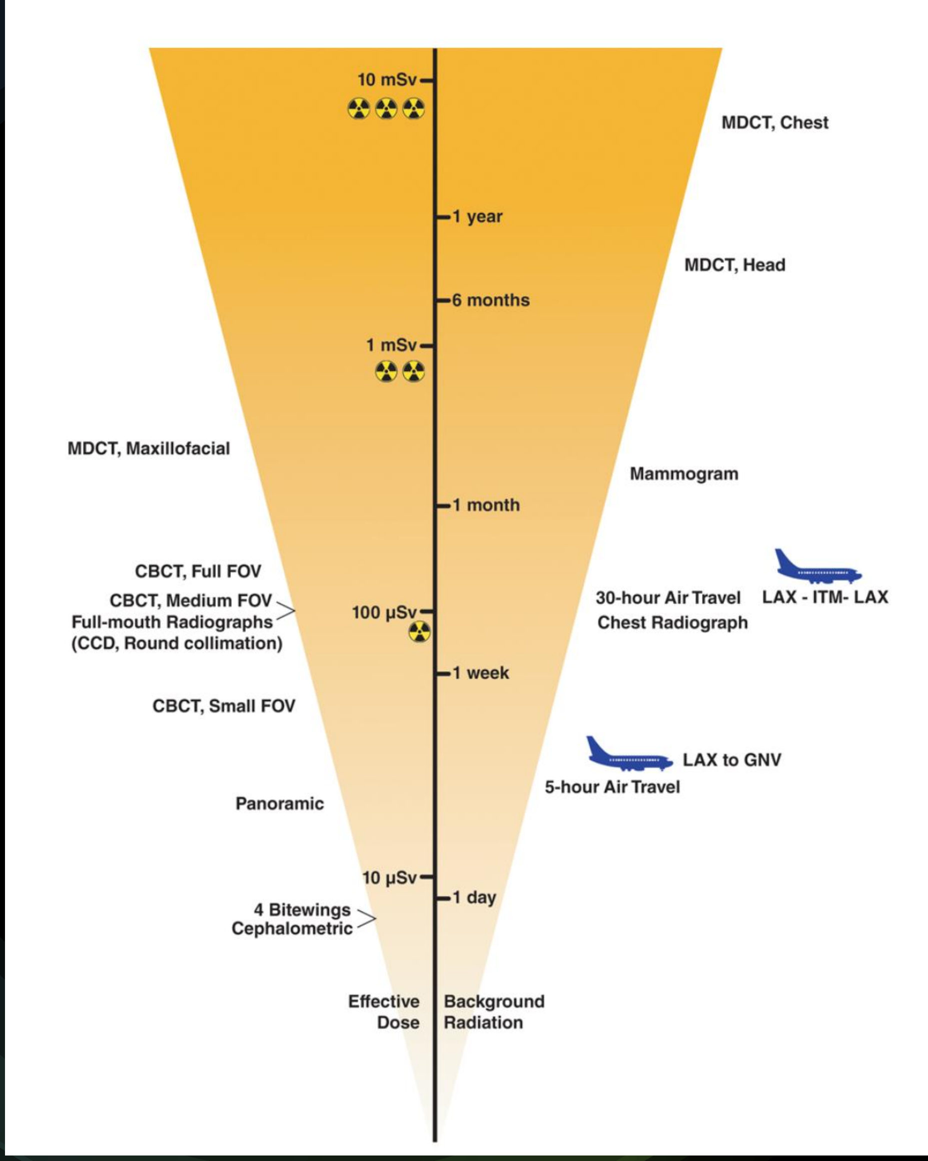
- Usually reported as effective dose, a measure of the stochastic risks
- These effective doses (ED) consider the absorbed doses to various organs from a specific radiographic procedure, and the relative sensitivity of the exposed organ to stochastic effects of radiation
- ED does not consider age, gender, and individual susceptibility factors - not intended to represent an individual's radiation dose from that procedure
- A convenient measure to compare the relative risks from different radiographic examinations and to convey the relative magnitude of the risk to patients

Typical Effective Dose From Radiographic Examinations

Examination	Median Effective Dose	Equivalent Background Exposure[a]
Intraoral[b]		
Rectangular collimation		
Posterior bite-wings: PSP or F-speed film	5 µSv	0.6 day
Full-mouth: PSP or F-speed film	40 µSv	5 days
Full-mouth: CCD sensor (estimated)	20 µSv	2.5 days
Round collimation		
Full-mouth: D-speed film	400 µSv	48 days
Full-mouth: PSP or F-speed film	200 µSv	24 days
Full-mouth: CCD sensor (estimated)	100 µSv	12 days
Extraoral		
Panoramic[b]	20 µSv	2.5 days
Cephalometric[b]	5 µSv	0.6 day
Chest[c]	100 µSv	12 days
Cone beam CT[b]		
Small field of view (<6 cm)	50 µSv	6 days
Medium field of view (dentoalveolar, full arch)	100 µSv	12 days
Large field of view (craniofacial)	120 µSv	15 days
Multidetector CT		
Maxillofacial[b]	650 µSv	2 months
Head[c]	2 mSv	8 months
Chest[c]	7 mSv	2 years
Abdomen and pelvis, with and without contrast[c]	20 mSv	7 years

Radiation Doses From Common Dental and Medical Radiographic Examinations

- The central axis (log scale) shows the effective dose and the equivalent period of background radiation exposure
- The radiation symbols adjacent to the dose identify the relative radiation level categories



Relative Radiation Level Designations, American College of Radiology

Relative Radiation Level	Adult Effective Dose Range	Pediatric Effective Dose Range
☢	<100 μSv	<30 μSv
☢☢	100 μSv to 1 mSv	30 μSv to 300 μSv
☢☢☢	1–10 mSv	300 μSv–3 mSv
☢☢☢☢	10–30 mSv	3–10 mSv
☢☢☢☢☢	30–100 mSv	10–30 mSv

From ACR Appropriateness Criteria, Radiation Dose Assessment, https://www.acr.org/Quality- Safety/Appropriateness-Criteria

Communicating radiation risks to patients

- Allow the patient to fully express their thoughts. Do not interrupt the patient's comments or belittle the patient's concerns. Acknowledge their concerns and indicate that you understand their apprehension.
- Inform the patient why you need radiographs as part of their personal diagnosis. Assure new patients that you will contact their previous dentist to obtain previous radiographs that may assist you in their diagnosis without unnecessarily re-exposing the patient.
- Emphasize that you will make only the exposures you specifically need for the patient's benefit guided by their symptoms and your clinical evaluation and not by a predetermined protocol.
- Educate and reassure the patient that you make consistent efforts to minimize their radiation dose by describing the many measures you take to reduce patient exposure. These may include using digital sensors or fast-speed film, rectangular collimation, and appropriate selection criteria.
- Convey the relative magnitude of the dose that the patient is likely to receive. To help the patient place the risk in perspective, compare the magnitude of these risks with other sources of radiation exposure, such as an airline flight or other common imaging procedures. Note that dentomaxillofacial radiography delivers negligible to minimal doses, relative to other medical imaging procedures.

Implementing Radiation Protection

Regulatory agencies and bodies

- ICRP is an independent, international organization that develops and disseminates recommendations and guidance on protection against ionizing radiation. Its recommendations are adopted or modified to establish radiation regulatory requirements in several countries worldwide.
- In the United States, NCRP, chartered by the US Congress, formulates guidance and recommendations on radiation protection and radiation measurements. The NCRP-recommended dose limits are incorporated into federal regulations.
- The current occupational exposure limits have been established to ensure that no individuals will have deterministic effects and that the probability for stochastic effects is as low as reasonably and economically feasible. Globally, it is important for dental personnel to be familiar with regulations and recommendations from their local and national radiation protection agencies.

Implementing Radiation Protection

Guiding Principle

1. Justification: situations where the benefit to a patient from the diagnostic exposure likely exceeds the risk of harm

2. Optimization: dentists should use every reasonable means to reduce unnecessary exposure to their patients, their staff, and themselves
   ALARA (*A*s *L*ow *A*s *R*easonably *A*chievable)

3. Dose limitation: provides dose limits for occupational and public exposures to ensure that no individuals are exposed to unacceptably high doses

Means for Reducing X-Ray Exposure

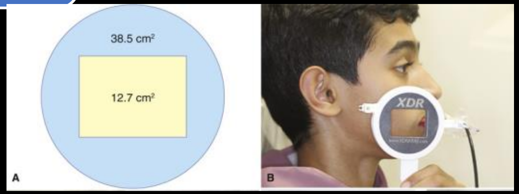
Use Good Clinical Judgment and Apply Evidence-Based Imaging Guidelines

- Make radiographs when they are likely to contribute to diagnosis and treatment planning
- Use selection criteria to assist in determining type and frequency of radiographic examinations

Means for Reducing X-Ray Exposure

Use Best Practices in Radiographic Imaging- Patient Protection

- Optimize your exposure settings
- Intraoral radiography
  - E/F-speed film or digital sensors
  - Holders to support film/digital sensors
  - Rectangular collimation: radiation dose up to 5-fold
  - Make exposures with 60–70 kVp
  - Avoid retakes
- Panoramic radiography
  - Rare-earth screens for film imaging
  - use digital systems



66% less than the round collimation field

Means for Reducing X-Ray Exposure

Use Best Practices in Radiographic Imaging-Patient/ Personnel Protection

Cephalometric radiography

- Rare-earth screens for film imaging
- Use digital systems

Cone beam computed tomography (CBCT)

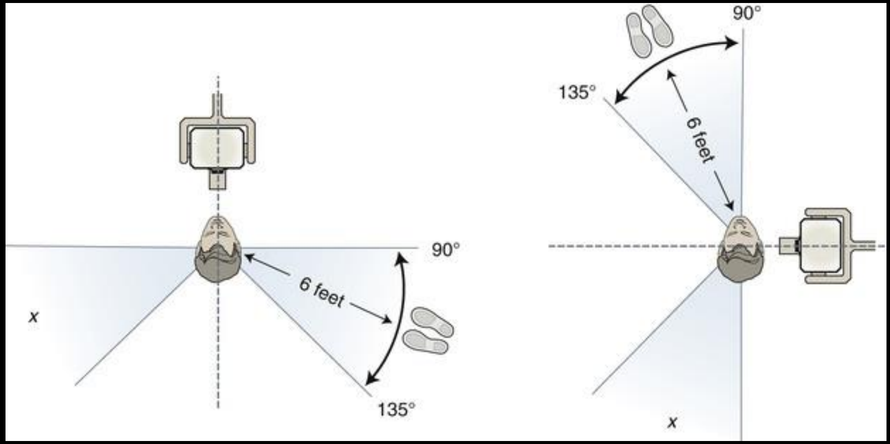
- Restrict the field of view to cover the region of interest

Film-based imaging

- Use time-temperature processing rather than “sight” processing
- Use an automatic processor

Means for Reducing X-Ray Exposure

Use Best Practices in Personnel Protection



- Barriers
  - Position-and-distance rule: Stand at least 6 feet (2 m) from the patient, at an angle of 90 to 135 degrees to the central ray of the x-ray beam
  - Never hold films or sensors in place
  - Qualified expert to design and construct dental offices
  - Walls must be of sufficient density or thickness so that the exposure to these nonoccupationally exposed individuals is less than 0.02 mGy/week (1 mGy/year)
- For handheld devices,
  - Ensure the protective backscatter shield is in place
  - Hold in a horizontal position, oriented perpendicular to the operator

Means for Reducing X-Ray Exposure

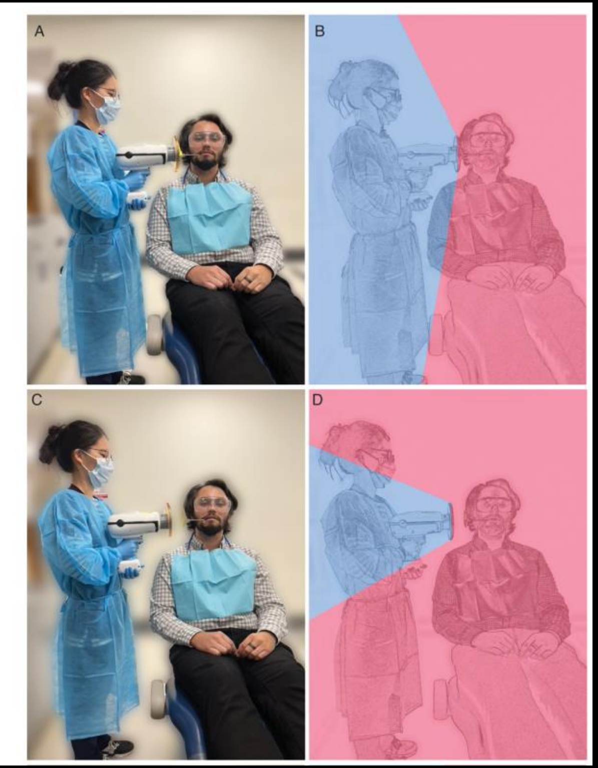
Handheld radiographic devices

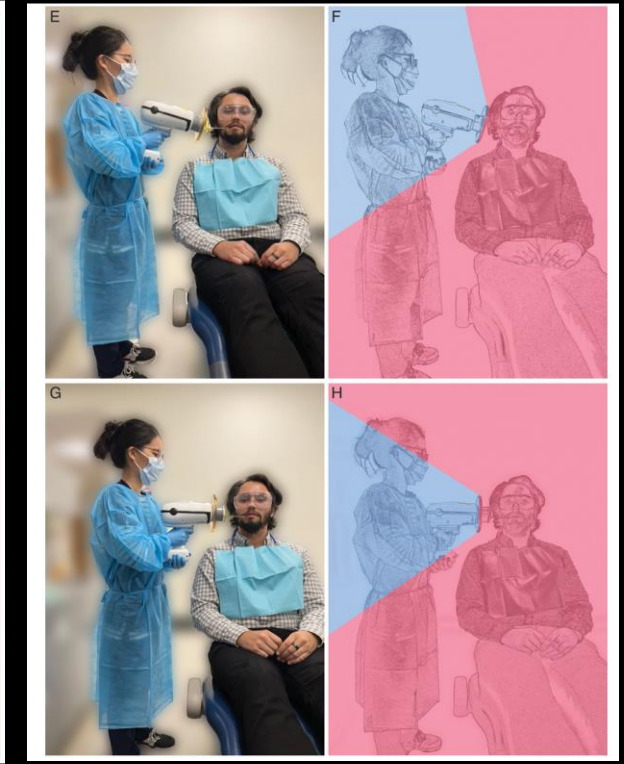
Safe operation of handheld dental x-ray devices.

The regions shaded in blue represent the area where scatter radiation is blocked by the shield. (A and B) Proper operation.
The device is held horizontal with the collimator ring placed close to the patient's skin surface. (C and D) Improper handling.

The device is held back away from the patient. (E and F) Improper handling.

The device is held at an angle and the shield is not vertical. (G and H) Improper handling. The protective shield is slid back from the edge of the collimator ring.





Means for Reducing X-Ray Exposure

Use Best Practices in Personnel Protection- Personnel Monitoring Devices

- Several companies offer dosimetry-monitoring services
- Optically stimulated luminescent dosimeter (OSLD)-widely used technology

  strip of crystalline aluminum oxide (Al2O3:C) luminesces in proportion to the amount of radiation exposure
- Dosimeter with a USB interface allowing the end user to directly perform the dose reading via the internet--another technology
- Both these technologies are sensitive to a dose as low as 10 µSv of x- rays.
- The ADA recommends that workers who may receive an annual dose greater than 1 mSv should wear personal dosimeters to monitor their exposure levels.
- Pregnant dental personnel operating x-ray equipment should use personal dosimeters, regardless of anticipated exposure levels (ADA 2012).

Considerations for the Pregnant Patient/Personnel

Identify risk

	Threshold Dose	Risk From Oral and Maxillofacial Imaging
Tissue Reaction Prenatal death Microcephaly Growth retardation Intellectual disability	100 mGy	None, fetal dose is approximately 10,000-fold lower than threshold
Stochastic Effect Radiation-induced cancer	In utero, not observed <10 mGy	Negligible, approximately 1 in 1.7 million

Recommendations for pregnant patients

- Lead aprons are no longer required due to negligible doses to the embryo/fetus during dentomaxillofacial radiology. The dosage that is received is from internal scatter radiation, and a lead apron does not offer protection against this.
- Verification of pregnancy status no longer required.
- All guiding principles of radiation protection and patient dose reduction measures must be strictly adhered to.

Recommendations in pregnant personnel

- US Code of Federal Regulations requires monitoring if the dose to the embryo/fetus is likely to be >1 mSv.

Dose limits

Recommended Dose Limits for Human Exposure to Ionizing Radiation

	NCRP[a]	ICRP[b]
Occupational Exposure		
Annual effective dose	50 mSv/year	20 mSv, averaged over defined 5-year periods
Cumulative effective dose	10 mSv × age	100 mSv in 5 years *and* should not exceed 50 mSv[c] in any single year
Annual equivalent dose		
Lens of eye	Absorbed dose of 50 mGy	20 mSv, averaged over defined 5-year periods, *and* exposure in any single year should not exceed 50 mSv
Skin, hands, and feet	500 mSv	500 mSv
Pregnant workers	0.5 mSv/month to embryo/fetus	1 mSv to the embryo/fetus after declaration of pregnancy
Public Exposure[d]		
Annual effective dose	1 mSv (continuous or frequent exposure) 5 mSv (infrequent exposure)	1 mSv
Annual equivalent dose in:		
Lens of eye	15 mSv	15 mSv
Skin	50 mSv	50 mSv

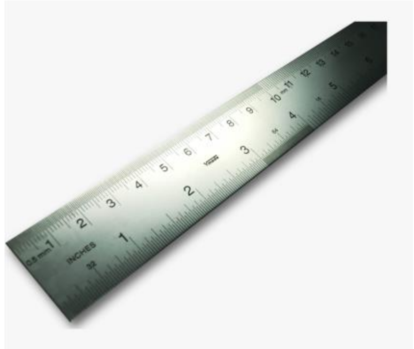
National Council on Radiation Protection and Measurement
International Commission on Radiological Protection

- Occupational dose limits: Dentists and their staff who make diagnostic radiographs are considered as occupationally exposed individuals and thus should be familiar with regulatory dose limits in their region or country. The ICRP-established dose limit for occupationally exposed individuals is 20 mSv of whole-body radiation exposure. Although this level of exposure is considered to present only a minimal risk, every effort should be made to keep the radiation dose to all individuals as low as practical. The dental profession does well in limiting occupational exposure. The average dose for individuals occupationally exposed in the operation of dental x-ray equipment is 0.2 mSv—1% of the allowable dose limit.
- Public dose limits: Support staff members (e.g., receptionists and auxiliary staff who do not perform radiography) and patients are subject to dose limits for members of the general public. The recommended limits for exposure to the public do not include natural background radiation and radiation received for individual medical and dental care. There are no limits on the exposure a patient can receive from diagnostic examinations, interventional procedures, or radiation therapy; this is because these exposures are made intentionally for the direct benefit of the recipient. Individual circumstances make the setting of limits inappropriate.
- Increasing concern for minimizing patient exposure has led multiple institutions, including the NCRP, to issue diagnostic reference levels (DRLs) for medical and dental diagnostic imaging. DRL exposure values represent the acceptable upper limits for patient exposure (75th percentile of general practice), whereas achievable doses represent the median dose (50th percentile) in general practice. The NCRP recommends a DRL of 1.6 mGy entrance skin dose for intraoral periapical and bite-wing radiography. The NCRP further recommends an achievable dose of 1.2 mGy for intraoral radiography.

Government regulations

- Federal
  - OSHA(Occupational Safety Health Administration)-Staff
  - FDA(Food and Drug Administration)NEXT surveys of dental radiation exposures
  - NCRP (Nat Council of Radiation Protection)-#160
- State
  - Regulations, equipment registration, inspection

Minimize Dose by Good Practices



- TIME-reduce time of exposure
- DISTANCE-increase distance
- SHIELDING-use shielding